E1: Orders of magnitude

Week 5. Due Wed 10 Feb 23:59, reviewed Thu 11 Feb

About 2 hours. Submit one PDF (handwritten and scanned is fine) or a notebook on Canvas. You may use an LLM, but be ready to explain any answer at the board on Thursday.

1. Nelson

Nelson, *Biological Physics*: Your Turn 1A, Problems 1.3, 2.2 and 2.5.

2. The thermal energy scale

Calculate $k_B T$ at body temperature (37 °C) in J, eV, kJ/mol and pN · nm. Compare it with the energy of one ATP hydrolysis (about 20 $k_B T$ in the cell) and with a covalent C–C bond (about 350 kJ/mol). What does the comparison tell you about which structures in the cell are stable against thermal motion?

3. Python warm-up: fit a line with an uncertainty

Generate synthetic data $y = at + b +$ noise with $a = 2$, $b = 0.5$, 30 points for $t \in [0, 10]$ and Gaussian noise with standard deviation 1. Fit a straight line and report $a \pm \delta a$. Repeat 1000 times with new noise and compare the spread of the fitted a with the δa from a single fit. Plot one data set with the fit.

Hint: `np.polyfit(t, y, 1, cov=True)` returns the covariance matrix of the parameters.